



Supply Chain Performance Analysis

A comprehensive data analytics and machine learning study to identify operational inefficiencies, evaluate profitability trends, and generate actionable recommendations for supply chain optimization.

PYTHON · PANDAS · SCIKIT-LEARN · SEABORN

The Challenge: Operational Inefficiency at Scale



The organization faces delayed deliveries, inconsistent shipping performance, and fluctuating profitability across regions and product categories — eroding margins and customer trust.

Core Problem

Delayed deliveries and inconsistent shipping performance across regions

Approach

Data-driven analysis using EDA, statistical evaluation, and ML

Goal

Identify root causes and deliver actionable operational recommendations

Key Performance Indicators

A measurable framework was established to evaluate operational efficiency and flag areas requiring immediate attention.



Delivery Delay Rate

Percentage of delayed orders vs. total orders processed



Average Shipping Time

Average days from order placement to delivery completion



Profitability

Net profit generated across orders, regions, and product categories



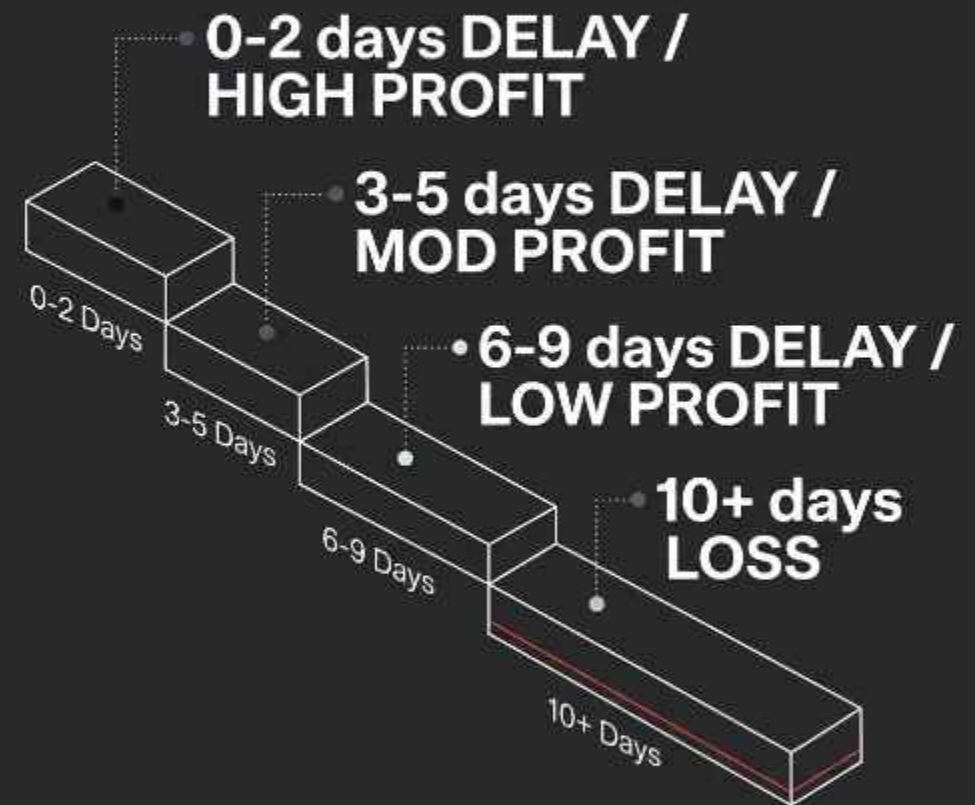
On-Time Delivery Rate

Orders delivered within expected timelines

Profitability Distribution & Delay Impact

What the Data Shows

Profit values were unevenly distributed — a significant portion of orders generated low or negative margins. Loss-making transactions were consistently associated with delayed deliveries and higher shipping costs.



→ High sales volume did not guarantee higher profitability

→ Delayed shipments incurred additional operational overhead

→ As delay days increased, profitability declined



BOTTLENECK DETECTION

Where Operations Break Down



Shipping Modes

Certain methods consistently experienced longer delivery times, especially during peak demand periods



Geographic Regions

Specific areas showed recurring delay patterns due to infrastructure limitations and logistics complexity



Order Processing

High order volumes during peak periods created warehouse overload and downstream delivery disruptions



Inventory Movement

Delays in inventory allocation and warehouse handling propagated inefficiencies across the supply chain

Five Root Causes of Delivery Delays



Key Insight

Delays are not caused by a single factor. They result from a **combination of logistics, operational planning, and demand management challenges** that compound over time.

- Addressing these root causes holistically — rather than in isolation — is essential for meaningful operational improvement.

Seasonal Patterns & Delay Frequency



What the Trends Reveal

Delay frequency spiked during high-demand periods, with certain months showing significantly higher operational pressure. Seasonal fluctuations directly impacted shipping timelines and warehouse throughput.

Peak Periods

Higher order concentration increased processing time and delay risk

Seasonal Impact

Recurring patterns enable proactive workforce and inventory planning

Predictive Modeling for Delivery Outcomes

ML models were trained to predict delivery-related outcomes and identify the most influential operational factors.



The predictive model successfully identified patterns associated with delayed deliveries. Top predictive features included **shipping mode, order region, delivery status, processing time, and product category** — enabling proactive risk identification and resource allocation.

Six Priorities for Operational Improvement



Improve Delivery Planning

Optimize routes with real-time monitoring and automated tracking



Strengthen Inventory Management

Use demand forecasting and maintain safety stock for high-velocity products



Optimize Shipping Methods

Evaluate underperforming modes and set partner performance metrics



Deploy Predictive Analytics

Build early-warning systems and real-time dashboards for operational risks



Enhance Regional Logistics

Strengthen support in delay-prone regions with localized distribution strategies



Improve Customer Experience

Increase shipment transparency and reduce delivery uncertainty

From Data to Actionable Intelligence

What This Analysis Proved

This project demonstrated that **delivery performance has a direct, measurable impact on profitability** and customer satisfaction. By applying Python-based analytics and machine learning, the organization can move from reactive problem-solving to proactive operational planning.

Project Workflow

01

Data Collection & Cleaning

02

Exploratory Data Analysis

03

Feature Engineering & Visualization

04

Statistical & ML Modeling

05

Business Recommendations

Delivery delays are predictable — and preventable

Historical data is a strategic asset for future optimization

Data-driven decisions outperform intuition-based planning